

Synthesis of α -Fe particles using a modified metal-membrane incorporation technique[☆]

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Abstract

We describe the synthesis of α -Fe particles using a modified metal-membrane incorporation technique based on diffusing metal ions through a dialysis membrane. The diffusion-time was varied up to 15 min. The samples were characterized using X-ray diffraction and scanning electron microscopy (SEM). The average diameter of the α -Fe crystallites was found to vary in the range 17–26 nm depending on the diffusion-time. The hysteresis loops were measured at room temperature while the zero-field-cooled (ZFC) and field-cooled (FC) magnetizations were measured down to 2 K. Both, the remanent magnetization and the coercive field were found to vary with the orientation of the applied magnetic field with respect to the membrane. An irreversible behavior in the ZFC and FC magnetization was also observed. The results are in good agreement with those expected for a three-dimensional (3D) system of nanoparticles with uniaxial anisotropy.

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1. Introduction

Over the last 40 years, BH_4^- reduction of metal ions has been used extensively for the formation of metallic particles and it is perhaps the most widely used method for producing ultra-fine materials [1]. In recent years, it gained additional importance due to the increasing interest in the magnetism of nanoparticle systems [2]. The recipes used for aqueous BH_4^- reductions of metal ions are so varied that it is startling. Almost any change in the procedure can make a difference. Concentrations, BH_4^-/M ion ratio, pH, method of mixing, and rate of mixing can determine particle size and reactivity and it can even change the product identity. In this paper, we describe the synthesis of α -Fe using a modified metal membrane incorporation technique. Instead of fixing the transition metal ions on the

polymer matrix [3], we allow the metal ions to diffuse through a dialysis membrane. This process seems to be faster and more efficient to prepare membranes impregnated with α -Fe nanoparticles. The morphology and the magnetic properties of these new magnetic films were also investigated.

2. Sample preparation and experimental techniques

The α -Fe films were prepared in a two-reservoir homemade cell. One of the reservoirs was filled with a 0.1 mol L^{-1} FeSO_4 solution while the other one was filled with 0.1 mol L^{-1} sodium borohydride dissolved in methanol and 0.1 M sodium hydroxide. The reservoirs were connected by dialysis membranes. The metal deposition was controlled by the time the solutions diffused through the membrane. After that, the membrane was removed from the cell, it was washed with distilled water and it was dried at room temperature. A 10 min diffusion-time is enough to grow a film such as the one shown in the SEM

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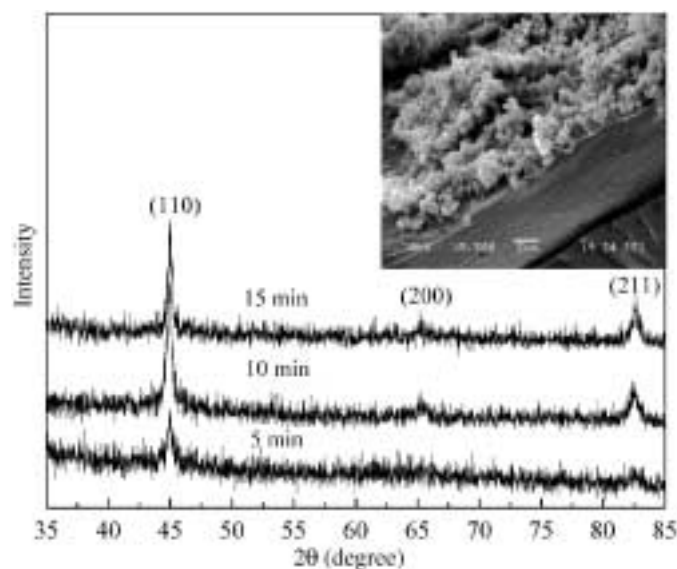


Fig. 1. XRD pattern for the 5, 10 and 15 min diffusion-time samples. The diffratograms are shifted on the vertical scale for clearness. The inset is a SEM image showing the α -Fe particles and the dialysis membrane for a 10 min diffusion-time sample.

Table 1
Summary of the magnetic parameters obtained with H applied parallel (perpendicular) to the membrane

Parameter	Samples		
	5 min	10 min	15 min
D (nm)	18.8	17.6	26.0
H_C (Oe)	66 (98)	64 (96)	33 (84)
M_R (emu/g)	11 (3)	22 (4)	16 (5)
M_S (emu/g)	45 (39)	67 (53)	76 (59)
K_{eff} (10^6 emu/cm ³)	0.9	1.2	2.4

image in Fig. 1. The structural properties of the samples were characterized by X-rays diffraction using the Cu-K α radiation and by scanning electron microscope (SEM). The magnetic properties were investigated using a SQUID and a vibrating sample magnetometers for applied magnetic fields H varying in the ± 14 kOe range.

3. Experimental results and discussions

Samples were obtained from the same start solution for 5, 10 and 15 min diffusion-time. The presence of α -Fe was detected by XRD in all the samples. The average size of the crystallites was obtained by the Rietveld refinement (see Table 1). The XRD patterns for the three membranes incorporated with α -Fe particles are shown in Fig. 1. The inset in Fig. 1 shows a SEM image for the same α -Fe particles-membrane system. The room temperature (RT) magnetization M curves were measured by varying the angle that H makes with the membrane. For instance, Fig. 2 shows M for H applied parallel (solid circles) and perpendicular (closed circles) to the membrane for a 15 min diffusion-time sample. The upper left inset shows the

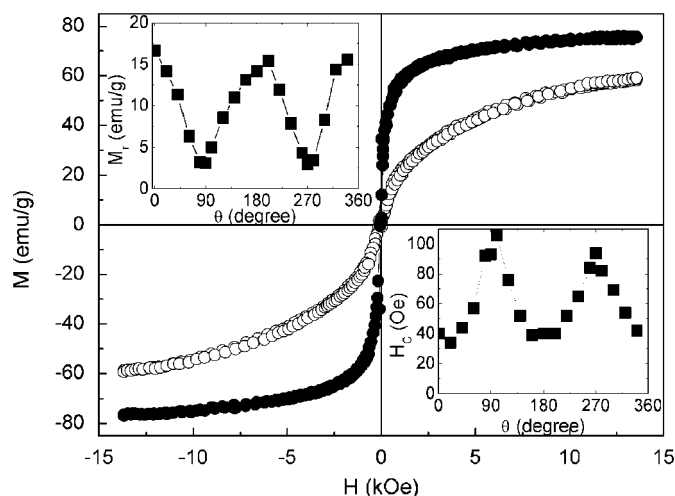


Fig. 2. RT hysteresis loops for a 15 min diffusion-time sample measured with H applied parallel (closed circles) and perpendicular (open circles) to the membrane (see the text for details about the insets).

remanent magnetization M_R vs. θ data while the lower right one shows the coercive field H_C vs. θ data where θ is the angle that H makes with the membrane. Such angular dependences result from the interplay between the demagnetizing field and the low magnetic anisotropy present in these materials, keeping most of the magnetic moments in the plane of the film. The M_R/M_S ratios ($= 0.24, 0.33$ and 0.21 for the 5, 10 and 15 min, respectively), on the other hand, are smaller than the theoretical value (0.5) obtained for an assembly of noninteracting three dimensional (3D) random particles system. Dipolar interaction and thermal agitation are expected to reduce the M_R/M_S ratio. Besides, our results are in good agreement with the values reported for α -Fe particles with dimensions close to the ones investigated in the present work [4]. An effective anisotropy K_{eff} can be estimated from the asymptotic behavior of the high field susceptibility data ($\partial M/\partial H \approx \alpha K_{eff}^2/M_S H^3$, where $\alpha = 0.533$) [5]. Both, the structural and magnetic parameters were found to depend on the diffusion-time. These parameters are summarized in Table 1. M was also measured using the zero-field-cooled (ZFC) and field-cooled (FC) procedures from 2 to 250 K with $H = 100$ Oe. An irreversible behavior determined by the ZFC and FC curves was observed with the curves crossing each other at a temperature close to 150 K. M did also decreased when the temperature is reduced. Further research is being done along this direction to better clarify these results.

4. Conclusions

In summary, a simple route was used to produce nanoparticles of α_0 Fe embedded in a dialysis membrane. It was shown that the structural and the magnetic properties of the particles can be tuned by the diffusion-time. Highly anisotropic and irreversible behaviors were signatures found in the magnetic properties of the films prepared by this new technique.

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